
Agent-Augmented Sequence Modeling for Explainable ICU Diagnosis Prediction

Midterm Report

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Abstract

This midterm report describes progress toward evaluating whether a multi-agent large language model (LLM) reasoning layer can improve primary ICD-10 diagnosis prediction accuracy over a trained LSTM baseline when applied at inference time to MIMIC-IV ICU data. We use the first 12 hours of ICU admission as input, predicting across 50 primary ICD-10 diagnostic categories for a cohort of 14,968 stays. Our implemented bidirectional LSTM baseline achieves 26.91% Top-1 accuracy and 72.87% Top-5 accuracy on the held-out test set. The 46-percentage-point gap between Top-1 and Top-5 accuracy constitutes the motivating ceiling for Phase 2: a multi-agent debate system that re-ranks the LSTM’s shortlist using structured clinical reasoning. The second half of the project will focus on implementing, evaluating, and ablating this agent reasoning layer.

1 Introduction and Project Overview

ICU patients represent the highest-acuity population in hospital medicine. Timely and accurate identification of a patient’s primary diagnosis within the first hours of admission directly informs treatment decisions, triage priority, and resource allocation. While clinical scoring systems such as APACHE and SAPS have long served as decision-support tools, they were designed for mortality and severity prediction rather than diagnostic classification. Electronic health records (EHRs) now provide a rich, temporally dense signal that machine learning models can exploit to produce early diagnostic predictions. Long Short-Term Memory (LSTM) models trained on ICU time-series have been shown to produce high-confidence predictions, but they lack explicit clinical reasoning, making errors difficult to detect or correct. As such, this project investigates whether the addition of a multi-agent LLM reasoning layer can improve primary ICD-10 diagnosis prediction accuracy on held-out MIMIC-IV ICU codes, as measured by Top-1 accuracy and AUROC.

This project involves a two-stage pipeline: a bidirectional LSTM trained on the first 12 hours of MIMIC-IV ICU time-series data to generate a ranked shortlist of ICD-10 diagnosis candidates, followed by a multi-agent LLM reasoning layer that re-ranks those candidates using structured

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clinical rationale. This report documents the completion of Stage 1 (data preprocessing and LSTM baseline) and outlines the design of Stage 2 (multi-agent debate system) for the second half of the course.

2 Background

Machine learning models trained on ICU EHR time-series data have demonstrated that clinically meaningful signals for diagnosis and risk prediction are present within the first hours of admission. Studies using large EHR benchmarks show that models trained on early physiological measurements can predict outcomes with performance that matches or exceeds traditional clinical scoring systems [1] [2].

Deep learning models like recurrent neural networks (RNNs) and Long Short-Term Memory (LSTM) architectures are shown to perform well on multivariate clinical time-series tasks [3] [4]. LSTM networks in particular are well suited to these tasks because they capture temporal dependencies across irregular measurement intervals—a structural property shared with ICU chartevents data [5] [6]. Prior work has applied LSTMs to ICU mortality prediction, length-of-stay estimation, and phenotyping tasks, establishing them as a natural baseline architecture for the diagnostic classification problem addressed here.

However, despite strong predictive performance, these models operate as black boxes: they produce high-confidence outputs without generating any clinical rationale, and their failure modes on ambiguous or rare presentations are difficult to detect or correct. Large language models (LLMs) have recently shown promise in medical reasoning and differential diagnosis generation by integrating clinical context, but they still remain unreliable when applied to structured physiological data [7] [8] [9] [10]. This gap suggests that LLMs are not standalone replacements for sequence models, but rather candidate augmentation layers that can operate on the model’s output.

This key limitation has motivated hybrid approaches that combine predictive models with reasoning systems to improve reliability in clinical settings. One such approach is MedGellan, which uses an LLM to generate reasoning-based clinical guidance from medical records, improving diagnostic performance compared to using raw inputs alone [11]. Multi-agent LLM systems further extend this by generating multiple candidate explanations and deciding among them to improve decision quality, but their use as inference-time reasoning layer on top of trained clinical prediction models remains largely unexplored [12]. Evaluating whether such a layer can enhance model outputs addresses a critical gap for healthcare and AI, testing a novel approach to improving robustness, interpretability, and clinical alignment without retraining existing systems.

3 Data

All data are drawn from MIMIC-IV, a large, freely accessible, de-identified EHR dataset from Beth Israel Deaconess Medical Center [2]. We restrict to ICU admissions only, using four source tables: `chartevents` (vital signs, ventilator settings, nursing assessments), `labevents` (laboratory results), `inputevents` (medication and fluid administrations), and `procedureevents` (ICU procedures such as intubation and dialysis). The prediction target is the primary ICD-10-CM discharge diagnosis code (sequence number = 1), filtered to the 50 most frequent codes in the cohort.

3.1 Cohort Construction and Inclusion Criteria

The final cohort consists of 14,968 ICU stays from 11,973 unique patients, subject to three inclusion criteria: (1) ICU stay duration sufficient to yield at least 12 complete hourly observation windows, (2) presence of at least one primary ICD-10 discharge code among the top-50 most frequent codes across all MIMIC-IV ICU stays, and (3) exclusion of neonatal admissions and stays with incomplete or corrupt `chartevents` records. Patient-level train/validation/test splitting (70/15/15) was applied so that no patient appears in more than one split (see Table 1).

Table 1: Patient-level data splits.

Split	Patients	ICU Stays	Percentage
Train	8381	10485	70%
Validation	1795	2268	15%
Test	1797	2215	15%
Total	11973	14968	100%

3.2 Feature Engineering

The observation window is fixed to the first 12 hours of ICU admission. For each hour and each clinical variable, we extract an interleaved value-mask pair: the recorded measurement value (or 0.0 if not recorded in that hour) and a binary indicator (1 = observed, 0 = missing). This design allows the model to distinguish true zero values from missingness.

Feature selection is data-driven: item IDs are ranked by frequency of occurrence across the cohort, and the top-K most frequent items are selected per source table (Table 2).

Table 2: Feature sources and dimensionality.

Source Table	Top-K Selected	Feature Pairs (val+mask)	Examples
chartevents	100	200	Heart rate, SpO2, GCS
labevents	50	100	Creatinine, lactate, potassium
inputevents	50	100	Vasopressors, IV fluids, sedatives
procedureevents	50	100	Intubation, dialysis, arterial line
Total	250	500 (+2 output cols = 502)	-

This yields 502 features per timestep (250 value-mask pairs from 4 source tables, plus 2 additional output columns). Item IDs are mapped to human-readable names using MIMIC-IV’s `d_items` and `d_labitems` dictionaries, with deduplication applied to resolve naming conflicts. Overall observation density is approximately 3.8% across all feature-hour pairs, reflecting the inherent sparsity of ICU data. All value columns are standardized using a `StandardScaler` fitted on the training split only, and mask columns (binary 0/1) are preserved as-is. In addition to the raw time-series, a structured `clinical_summary` dictionary is computed for each stay (aggregating mean, min, max, last value, and trend per observed feature) for downstream use by the debate agents in Stage 2.

4 Baseline Implementation

The baseline model is a stacked, bidirectional Long Short-Term Memory (BiLSTM) network with a fully-connected classification head. The input is a tensor of shape [12 timesteps $\times$ 502 features], representing the first 12 hours of ICU observation. The output is a probability distribution over 50 ICD-10 diagnostic classes. Bidirectionality allows the model to attend to both earlier and later timesteps when constructing its hidden representation, which is useful for detecting temporal patterns across the admission window.

4.1 Architecture

The architecture includes an input projection layer (Linear 502 $\rightarrow$ 256) that decouples the input width from the LSTM hidden size. This feeds into a bidirectional LSTM with 2 layers and a hidden size of 256. Because the forward and backward states are concatenated, the effective hidden dimension is 512, with a dropout rate of 0.3 applied between layers. A classification head then processes the output through `LayerNorm(256)`, followed by `Dropout(0.3)` and a final Linear(512 $\rightarrow$ 50) projection. The model totals approximately 2.7M trainable parameters.

4.2 Training Setup

The model is trained using cross-entropy loss with inverse-frequency class weights to partially compensate for severe label imbalance, which reaches up to 2,164:1 in the training data. Optimization is handled by *Adam* with a learning rate of 1e-3 and weight decay of 1e-5, paired with a *ReduceLROnPlateau* scheduler that monitors validation macro F1 with a patience of 3 and a reduction factor of 0.5. Training runs for a maximum of 60 epochs with a batch size of 32, gradient clipping at a max norm of 1.0, and early stopping triggered after 5 epochs without improvement in validation macro F1. The best checkpoint was saved at epoch 10, achieving a validation macro F1 of 15.30%.

4.3 Evaluation Protocol

Evaluation is performed on the held-out test set (2,215 stays). Primary metrics are Top-1 accuracy (exact match between predicted and true primary ICD-10 code) and Top-K accuracy for K = 1, K = 3, and K = 5. Secondary metrics are macro F1 and weighted F1. Macro F1 is used for early stopping during training because it is more sensitive to rare-class performance, and weighted F1 is reported alongside it to characterize performance on high-frequency diagnoses.

4.4 Baseline Results

As shown in Table 3, the LSTM baseline achieves 26.91% Top-1 accuracy and 72.87% Top-5 accuracy on the test set. The 46-percentage-point gap between Top-1 and Top-5 accuracy is the central finding of Stage 1: in approximately 73% of test cases, the correct diagnosis is present within the model’s top-5 candidate list, but is not ranked first. This ‘recoverable ceiling’ is the core motivation for the LLM reasoning layer in Phase 2, as the debate system does not need to surface new candidates, but rather needs to correctly re-rank a shortlist that already contains the ground-truth answer in the majority of cases.

Additionally, the 12-percentage-point gap between weighted F1 (27.80%) and macro F1 (15.61%) confirms that model performance is skewed toward high-frequency diagnoses (e.g., sepsis, heart failure) and degrades substantially on rarer classes. This is a known challenge given label imbalance ratios of up to 2,164:1 in the training set.

Table 3: LSTM baseline performance on held-out test set (n = 2,215 stays).

Metric	Value (Test Set)	Notes
Top-1 Accuracy	26.91%	Exact match, primary ICD-10
Top-3 Accuracy	60.95%	True code in top-3 candidates
Top-5 Accuracy	72.87%	True code in top-5 candidates
F1 Weighted	27.80%	Reflects common class performance
F1 Macro	15.61%	Sensitive to rare class performance
Test Samples	2215 stays	Held-out patient-level split

5 Proposed Method: Design and Implementation Status

Stage 1 (implemented, described above) produces a ranked shortlist of top-K ICD-10 candidates with associated confidence scores from the frozen LSTM checkpoint. Here, we describe Stage 2 (upcoming), which introduces a multi-agent LLM debate layer that operates on this shortlist at inference time without modifying the LSTM weights.

5.1 Multi-Agent Debate Architecture

The debate system assigns one specialist LLM agent per each of the top-5 candidate diagnoses (five agents instantiated per patient), with all agents powered by *Gemini Flash 2.5*. Each agent receives: (1) its assigned ICD-10 candidate code and clinical description, (2) the structured `clinical_summary` dictionary produced during preprocessing (aggregated vitals, labs, and intervention history from the 12-hour window), and optionally (3) the LSTM confidence score for its assigned candidate. Each agent then produces a structured clinical rationale arguing for or against its assigned diagnosis,

grounded in the provided clinical data. To further support evidence-based reasoning, each agent will leverage a retrieval-augmented generation (RAG) component to incorporate relevant, up-to-date clinical guidelines and reference knowledge, improving the grounding and reliability of its rationale.

A sixth ranking agent then receives all five rationales along with the LSTM confidence scores and synthesizes them into a final ranked prediction with an explainable justification. The ranking agent’s output constitutes the Stage 2 Top-1 prediction, which is compared against both the LSTM Top-1 prediction and the ground-truth ICD-10 code. The entire agent layer operates on the frozen `best_model.pt` checkpoint, so no retraining occurs in Stage 2.

The debate payload is already generated: `debate_inputs.json` (20.8 MB) contains LSTM top-5 candidate codes, confidence scores, and clinical summaries for all 2,215 test stays. The Stage 2 agent ceiling (fraction of test cases where the ground-truth code appears in the LSTM’s top-5 shortlist) is 56.0%, establishing the maximum achievable Top-1 accuracy for the debate system under top-5 retrieval.

5.2 Implementation Status

- **Completed:** MIMIC-IV preprocessing pipeline (`patient_cards_v4.json` format with `clinical_summary`), BiLSTM training and evaluation, debate payload generation (`debate_inputs.json`)
- **In progress:** Optimize data preprocessing and BiLSTM training, multi-agent prompt engineering; determining optimal agent input composition (with or without LSTM confidence scores to avoid anchoring bias)
- **Planned:** Full agent inference loop, ranking agent implementation, comparative evaluation against LSTM baseline, ablation studies

5.3 Planned Experiments Beyond the Baseline

Several experiments are planned to evaluate and ablate the multi-agent debate system beyond the baseline. The primary comparison will measure Top-1 accuracy of the debate system against the LSTM baseline on the 2,215-stay test set. A Top-K retrieval ablation will assess how agent performance varies when candidates are shortlisted to the top 3 versus top 5, and an agent input ablation will compare performance with and without LSTM confidence scores included in the prompt to assess any anchoring effect. Additionally, a confidence calibration analysis will examine whether the debate system improves predictions specifically on cases where the LSTM assigns high confidence to an incorrect diagnosis.

6 Challenges, Risks, and Mitigation

6.1 Severe Label Imbalance

The training set exhibits a maximum class imbalance ratio of 2,164:1 (most frequent vs. rarest ICD-10 code). This is reflected in the 12-percentage-point gap between weighted F1 (25.25%) and macro F1 (13.49%), confirming that the model substantially underperforms on rare diagnostic categories. We have partially addressed this with inverse-frequency class weighting in the cross-entropy loss, and by using macro F1 rather than accuracy as the early stopping criterion. For Stage 2, we are considering grouping codes that share prefixes into clinically meaningful groups as an additional mitigation before finalizing the LSTM checkpoint for agent evaluation.

6.2 Sparse Temporal Observations

Observation density across all feature-hour pairs is approximately 3.8%, meaning the vast majority of entries in the $[12 \times 502]$ input tensor are zero. While the value-mask interleaving design allows the model to distinguish true zeros from missingness, high sparsity still limits the signal available per patient. Two mitigations are planned to address this. First, feature selection by observation frequency would reduce the input dimensionality by dropping columns that are never or rarely observed across the dataset, retaining only features with sufficient coverage to provide meaningful signal. Second, mask-gated feature processing would explicitly condition the model’s use of a value column on the

corresponding mask column, preventing spurious gradients from propagating through unobserved measurements.

6.3 ICD-10 Discharge Codes as Proxy Labels

Primary ICD-10 codes are assigned retrospectively and are greatly influenced by billing and documentation practices, which means that they may not always reflect the most prominent clinical condition during the ICU admission window. This introduces label noise that is difficult to quantify or correct, so our mitigation is methodological: we will report both weighted and macro F1 to distinguish model behavior on well-documented common codes from behavior on noisier rare codes. We will also acknowledge this limitation when interpreting results.

6.4 LLM API Cost and Latency at Scale

Running multi-agent LLM inference across 2,215 test cases with 5 specialist agents and 1 ranking agent per case represents approximately 13,290 individual LLM calls. This introduces cost and latency constraints that require careful batching and prompt optimization. To address this, we plan to run the full agent evaluation on a sampled subset first to validate the pipeline before scaling to the full test set.

7 Updated Timeline and Plan to Final Report

- Week 8 (current): Finalize data preprocessing and LSTM optimization (mask-gated features, class-balanced sampling experiments); lock best LSTM checkpoint for agent evaluation
- Week 9: Implement specialist agent prompt templates; run pilot agent inference on 100-case sample; evaluate anchoring effect of including LSTM confidence scores
- Week 10 & 11: Full agent inference run on all 2,215 test cases; implement ranking agent; collect Stage 2 Top-1 accuracy results
- Week 12: Analysis, figure generation (training curves, confusion matrices, per-class performance bar charts, confidence calibration plots), and report writing
- Week 13: Final report submission and presentation preparation

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